

Little LiFePO4 Battery Charger

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This circuit of single-cell LiFePO4 (lithium iron phosphate) battery charger is based on an LM358 operational amplifier (op-amp) and a couple of inexpensive and easy-to-get components. It can be powered from any USB port or USB standard power supply adaptor. It does not use any difficult-to-handle surface mount device (SMD) or a miniscule chip.

LiFePO4 batteries are best known for their safety because of their extremely stable phosphate-based chemistry. Also, these newer type of lithium batteries are inherently non-combustible. Hence, these can withstand harsh conditions, be it freezing cold or searing heat. Even when subjected to hazardous events (collision or short-circuit), these do not explode or catch fire, thus significantly reducing the chance of any harm. The phosphate chemistry also provide a longer life to the batteries.

Single-cell LiFePO4 batteries are now within the reach of electronics hobbyists. These are also available pre-wired as LiFePO4 charger boards/modules from online vendors. Most such Chinese modules (1S/3.2V) are built around a 16-pin QFN chip—the TP5000 as shown in Fig. 1.

Circuit and working

Circuit diagram of the little LiFePO4 battery charger is shown in Fig. 2. The pnp transistor S8550 (T1) used in the circuit routes input DC supply to the battery through silicon diode 1N4007 (D1). Transistor T1 is controlled through LM358 (IC1) op-amp. The features of IC1 include low offset, common-mode input range to ground, and high differential input voltage capability. Green-colour LED1 acts as charging indicator while LED2 indicates a fully charged battery. LED3 works as power indicator but it also provides a fixed reference voltage to



Fig. 1: LiFePO4 1S battery and TP5000 charger module

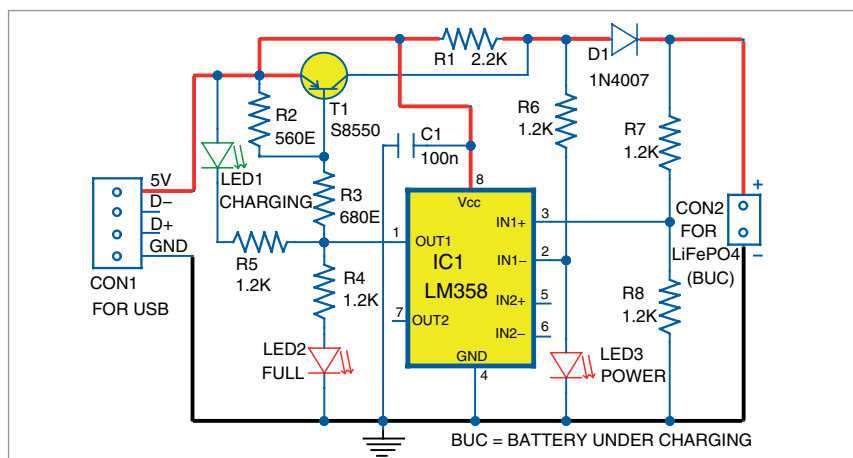


Fig. 2: Circuit diagram of the little LiFePO4 battery charger

the inverting input pin 2 of IC1.

A 1.2-kilo-ohm resistor network (R7 and R8) is connected across the battery terminals. Output from the junction of this resistor network goes to non-inverting input pin 3 of IC1. Since the circuit is designed as a single-cell LiFePO4 (3.2V) charger, it is configured to give the recommended output of 3.6V.

When the connected battery's

(BUC) voltage is less than the threshold voltage, transistor T1 starts charging the battery and LED1 lights up to show that charging process is in progress. When the LiFePO4 battery's voltage goes above a certain limit, charging current gradually decreases and finally T1 is switched off by IC1 to stop further charging. It happens just after the battery voltage reaches its expected 3.6V level, which is indi-